9 Radar and Fibre Optics

INTRODUCTION TO RADAR

RADAR stands for Radio Detection and Ranging. Some basic points related to RADAR are as follows:

i) The electromagnetic waves ranging in frequencies between 3kHz to 300GHz are referred to as radio waves. It has a wide range, spanning from 1 mm to 100 km.

ii) It is an object detection system that detects and determines the range (distance), and maps the objects such as aircraft, ships, guided missiles etc.

iii) The reflector is easily located and identified by observing the reflected signals termed as echoes.

iv) It can operate and measure distances almost in all the weather conditions, such as haze, fog, darkness, rain, and snow.

Grey Matter Alert!

A German physicist Heinrich Rudolf Hertz discovered the existence of electromagnetic waves in the 1880s at Karlsruhe Institute of Technology which was predicted by James Clerk Maxwell's. These electromagnetic waves were referred to as Hertzian and later named as radio waves.

PRINCIPLE OF RADAR

The operating principle of radar is the same as the principle of sound-wave reflection. If we shout in the direction of a sound reflection object such as a wall or hill, we will hear an echo.

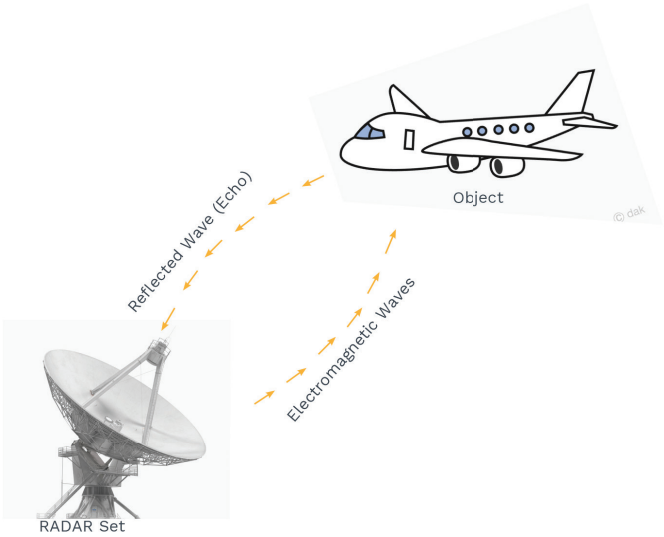


Fig. 9.1 Basic Principle of Radar

Definitions

"A remote sensing technique which is capable of gathering the information about the objects that are located at remote distances from the sensing device is termed as RADAR (Radio Detection and Ranging)".

The electromagnetic energy pulses which are used by the RADAR are shown in the above figure. The radio frequency (f_R) energy is transmitted, and then it is reflected from the reflecting object. This small portion of the energy which is being reflected and returned to the radar set (Radar receiver) is called an **ECHO**. This echo is used to obtain the distance of the reflecting object.
The distance can be calculated as:

$$R = \frac{c \times T_R}{2}$$

Where

T_R = Time taken by the wave to return to the RADAR receiver.

c = Velocity of the electromagnetic wave (velocity of light).

R = Distance between the RADAR set and the reflecting object.

Here in the above calculation, the object is assumed to be stationary.
The factor '2' in the denominator denotes the two-way propagation of the radio wave (one for transmission and the other for the reception).
For the moving objects, the calculation of the Radar Range (distance) will be discussed later.

Grey Matter Alert!

Basic working principle of a Radar system is that it sends the radio waves (electromagnetic waves) to the distant objects and analyse the reflected wave (echo signal) to gather the information about the object.

Rack Your Brain

How much range (in kilometres) can be covered by a Radar?

BLOCK DIAGRAM OF RADAR

The generalised block diagram of the basic Radar set is shown below.

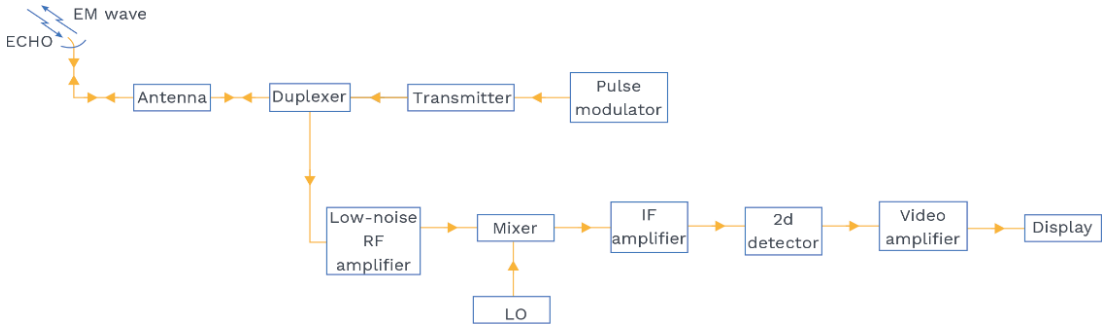


Fig. 9.2 Block Diagram of Basic Radar

Transmitter and modulator:
A transmitter can be an oscillator which is turned ON and OFF (pulsed) by the pulse modulator for generating a repetitive train of pulses.

Antenna:
It converts the electrical signals into electromagnetic waves for radiating into the space. For both transmitting and receiving purposes, a single antenna is used.

Duplexer:
It acts as a dual transmit/receive switch and protects the receiver from the high power of the transmitter.

Receiver:
The receiver is of the superheterodyne type. To reduce the noise contribution of the mixer, its first stage called the front end might be a low noise RF amplifier like a parametric amplifier or a low noise FET made from gallium arsenide is being used as a GaAs FET preamplifier. In many Radar applications, the front-end pre-amplifiers may be excluded.
As an input stage, the mixer has a greater dynamic range, less vulnerability to external electromagnetic interference, and less susceptibility to overload, when compared to a front-end RF pre-amplifier. The mixer and the local oscillator converts the RF signal to an intermediate frequency or IF.

IF amplifier:
The IF amplifier amplifies the mixer output at the intermediate frequency without producing appreciable distortions in the pulse waveform of the IF signal.

Grey Matter Alert!

Radar can easily sense the objects in the conditions such as: darkness, rain, haze and snow which is impossible for a human eyes.



Rack Your Brain

What is the operating frequency range of a Radar?



Rack Your Brain

What is the range of centre frequency and bandwidth for a typical IF amplifier?

The bandwidth of the IF amplifier should ideally be very wide for no envelope distortions of the IF signal. However, the inherent receiver noise accepted within such a wide IF bandwidth will reduce the signal-to-noise ratio of the echo signal. The bandwidth is made as small as possible until the waveform distortion becomes appreciable.
If the range of the radar is to be maximised, the peak amplitude of the echo signal must be maximised, and some waveform distortion may be tolerated. Accordingly, the bandwidth can be further minimised.
On the other hand, if a high precision of range measurement is desired, a steep rise of the pulse waveform is useful for which a larger IF bandwidth will be essential, although it will be inevitable to reduce the signal to the noise ratio. The centre frequency of a typical IF amplifier lies in the range of 30-60 MHz, and the bandwidth is of the order of 1 MHz.

Detector:
The envelope of the IF signal is extracted by an envelope detector. A diode detector may be used for this purpose. However, since the shape of the pulse envelopes is important to reduce distortion in the pulse waveform, the detector load is frequency compensated by inductance using the shunt peaking and series peaking techniques as used in the video detectors of the TV systems.

Video amplifier:
In this stage, the detected echo signal is amplified to a level where it can be properly displayed on a cathode ray tube screen. In this stage too, the waveform distortion is minimised by video compensation techniques.

Grey Matter Alert!

Limitations of a Radar are:
i) It cannot recognise the colour of the target.
ii) It cannot identify the internal aspects of the target.
iii) It can be interfered by several objects and mediums in the air.
iv) It cannot resolve multiple targets.

Displays:

The simplest form of display is called the A-scope, where the echo amplitude produces a deflection of the electron beam in the C.R.O. in the y-direction. The A-Scope is a deflection modulated display, and it is more suited for tracking applications.

Another type of display is the B-scope display which utilises rectangular coordinates to display range and the elevation angle. The echo in this display modulates the intensity modulation of the spot on the screen.

The most widely used form of display is perhaps, the plane position indicator abbreviated as PPI. This effectively maps, in polar coordinates, the plane position or the location of the target in azimuth and range. The PPI display is also intensity-modulated in which the echo modulates the electron beam like that in B-scope. In PPI, the electron beam rotates in synchronism with that of the axis of the main beam of the antenna pattern. In most radars, the PPI display is supplemented by an alternative display that is similar to the B-scope, but the elevation angle is replaced by the height of the target. This is called the range height display, abbreviated as RH display.

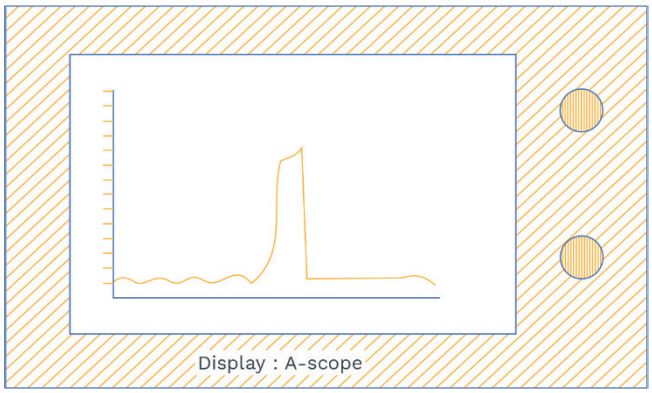


Fig. 9.3 A Typical RADAR Display

In a radar system, the detection and ranging part is supported by timing the delay between the transmission of a pulse and its reception. If the time delay is represented by Δt, then the range of radar can be obtained by:

$$R = c \times \left(\frac{\Delta t}{2} \right)$$

Where $c = 3 \times 10^8 m/s$

The factor of 2 in the formula comes because the radar pulses travel twice the range.

Definitions

"The time interval between the successive clock pulses is referred to as Pulse Repetition Time (PRT)".

Example 1:

Calculate the range of target if the time taken by the radar signal to travel to the target and back is 80 μsec.

Sol:

The correct answer is (12 km).

Given,

$$\Delta t = 80 \mu \, sec$$

$$\text{Range (R)} = c \times \left(\frac{\Delta t}{2} \right)$$

$$c = 3 \times 10^8 m / s$$

By putting the values in the equation, we get:

$$R = \left(3 \times 10^8 \right) \times \left(\frac{80 \times 10^{-6}}{2} \right)$$

$$R = 12km$$

Hence, the correct answer is (12 km).

COMMON PARAMETERS OF RADAR PULSE

Pulse Width (PW):

It is the duration of the pulse and has the units of time, i.e., sec or msec. But commonly, it is expressed in msec.

Definitions

"The number of Radar pulses transmitted per second is referred to as Pulse Repetition Frequency (PRF)".

Rest Time (RT):
It is the time interval between the pulses, and it basically denotes the time period during which the transmitter is not firing and the receiver waits for the reflected signal. It is also expressed in msec.

Pulse Repetition Time (PRT):
It is defined as the time interval between the start of the consecutive pulses. The unit of PRT is in time and expressed in msec.
Pulse Repetition Time = Pulse width + Rest Time

Pulse Repetition Frequency (PRF):
"The number of pulses transmitted per second is referred to as pulse repetition frequency (PRF)". It has a unit of time^{-1}, and it is expressed in Hz, i.e., (1 Hz = 1/s) or pulses per second (PPS). PRF is equal to the inverse of PRT, as shown below.

$$PRF = \frac{1}{PRT}$$

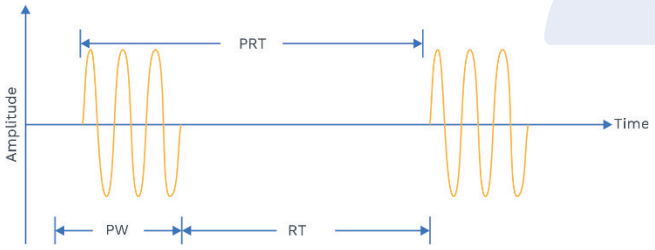


Fig. 9.4 Common parameters of a radar pulse

Rack Your Brain

How is pulse repetition frequency related to pulse repetition time?

Radio Frequency (RF):
"The frequency of the carrier wave is termed as radio frequency". A radio frequency is modulated to obtain the pulse train. It is represented in GHz or MHz.

Peak Power (P_t):
"The power averaged over that carrier frequency cycle, which occurs at the maximum of the pulse power is referred to as peak power".

Average Power (P_{av}):
"The average transmitted power over the pulse repetition time is referred to as average power".

$$P_{av}\left(\text{Average Power}\right) = P_t \times \frac{PW}{PRT} = P_t \times PW \times PRF$$

Duty Cycle:
"The ratio of average power to the peak power Is referred to as duty cycle."

or

The ratio of pulse width to the PRT.

or

Pulse width multiplied by PRF.
It may be written as,
Duty cycle = PW × PRF

Example 2:
If the transmitted peak power of radar is 100 KW, pulse repetition frequency is 1000 PPS, and the pulse width is 2 μsec, then calculate the average power in dbs.

Sol:
The correct answer is (23 dbs).
Given,

$P_t = 100\,kW$

$PRF = 1000\,pps$

$PW = 2\mu s$

We know that Duty Cycle = PW × PRF
By putting the value in the above equation, we get,

$\text{Duty cycle} = (2 \times 10^{-6}) \times (1000) = 0.002$

We know, the average power

$= P_t \times \text{Duty cycle} = \left(100 \times 10^3\right) \times 0.002$

$= P_t \times \text{Duty cycle}$

$= (100 \times 10^3) \times 0.002$

Grey Matter Alert!

The different types of Radar are: Bistatic Radar, Doppler Radar, Mono-pulse Radar, Passive Radar, Instrumentation Radar, Weather Radar, Mapping Radar, Navigational Radar, Pulsed Radar.

Definitions

"The range beyond which the target appears as second-time-around echoes are defined as the maximum unambiguous range".

Average power = 200 W
We can write in dbs, 200 W = 23 dbs
Hence, the correct answer is (23 dbs).

Example 3:
Determine the maximum unambiguous range and range resolution of a Pulse Radar having a pulse width is 5 μs at a PRF of 1500 Hz.

Sol:
The correct answer is (100 km) and (750 m).
Given,
PW = 5 μs

f_r = PRF = 1500Hz

$$R_{max}(unamb) = \frac{c}{(2 \times f_r)} = \frac{3 \times 10^8}{(2 \times 1500)} = 100km$$

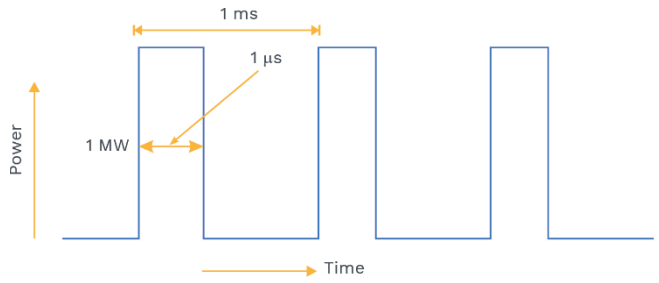
$R_{max}(unamb) = 100km$

Range Resolution

$$= c \times \frac{T}{2} = \frac{\left(3 \times 10^8\right) \times \left(5 \times 10^{-6}\right)}{2}$$

Range resolution = 7.5 × 100 = 750 m
Range Resolution = 750 m
Hence, the correct answer is (100 km) and (750 m).

Example 4:
A typical pulse waveform of radar is shown below figure. In which some parameters of radar are shown. Calculate the (a) average power, (b) duty cycle, and (c) maximum range of the radar.



Rack Your Brain

In a Radar system, the gain of the transmitting antenna is:
a) equal to the gain of receiving antenna
b) is less than the gain of receiving antenna
c) is greater than the gain of receiving antenna
d) none of the above

Definitions

"The distance between the Radar and a target is referred to as range of the target. It is denoted by R".

Sol:

From the figure, we have

Peak power $(P_t) = 1MW$

Pulse width $(T) = 1\mu s$

Pulse repetition period $(T_r) = 1ms$

We know, $PRF(f_r) = \frac{1}{T_r} = 1/(1\times10^{-3}) = 1000$

a) Average power

$P_{av} = P_t \times T \times f_r$

$P_{av} = (10^6)\times 1\times 10^{-6}\times(1000)$

$P_{av} = 1000\ W = 1\ KW$ (Ans)

b) Duty cycle $= T \times f_r$

$= (10^{-6})\times(1000) = 0.001$

$= 0.001$

c) Maximum unambiguous range

$$(R_{max}) = \frac{c}{(2\times f_r)} = \frac{(3\times10^8)}{(2\times1000)} = \left(\frac{3}{2}\right)\times10^5$$

$R_{max} = 150\ km$

$$R_{max} = \frac{150\times10^3}{1852} = 81nmi$$

$(1\ nmi = 1852\ m)$

Grey Matter Alert!

The commonly used display used in a Radar system is Plan Position Indicator (PPI). PPI radar display helps in providing the location and the direction of a targeted object on a map like presentation which is easy to interpret.

Example 5:

If the peak power is 100 kW, the pulse width is 3 μs and the rest time is 3000 μs, then obtain the average power.

Sol:

The correct answer is (100 W).

Given,

$P_t = 100kW$

PW=3μsec

RT=3000μsec

$P_{av} = P_t \times$ duty cycle

$$\text{Duty cycle} = \frac{PW}{PRT} \qquad [\text{Since PRT = PW + RT}]$$

$$\text{Duty cycle} = \frac{(3 \times 10^{-6})}{(3 \times 10^{-6} + 3000 \times 10^{-6})}$$

$$\text{Duty cycle} = 0.00099$$

Or $\text{Duty cycle} \approx 10 \times 10^{-4}$

$$P_{av} = (100 \times 10^{3}) \times (10 \times 10^{-4})$$

$$P_{av} = 100W$$

Hence, the correct answer is (100 W).

Rack Your Brain

What are the civilian applications of a Radar?

PULSE RADAR

To transmit a high-frequency impulsive signal of high power, pulse radar is used. By using the measured antenna position and the propagation time of the pulse signal, the distance, direction, and altitude of the target can be easily obtained.

A detailed block diagram of a high-power pulsed radar set is shown below.

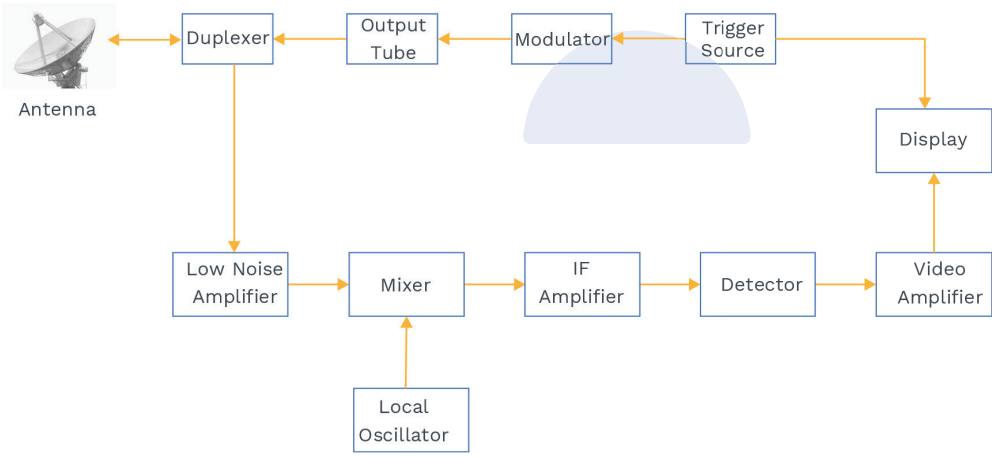


Fig. 9.5 Block Diagram of Pulse Radar

Duplexer:

It is basically a switch which connects the transmitter and receiver to the antenna on an alternate basis. Due to the high-power output of the transmitter, a duplexer is used to protect the receiver.

If the power transmitted is low, then a duplexer is not required.

Antenna:
An antenna is basically a transducer which converts the electrical signals into electromagnetic waves and transmit them in the desired direction. A synchro-transmitter is used in an antenna to keep the track of its orientation. Now a days, antenna systems are steered electronically.

Transmitter (Output tube):
The transmitter generates the radio waves to be sent and modulates them to form the pulse train. To provide a suitable range, the transmitter should amplify the signal to a high-power level. For providing the carrier wave, a Klystron, a Travelling Wave Tube (TWT) or a Magnetron can be used.

Radar modulator:
During the transmission period, to provide high power to the transmitter tube, a radar modulator is used.
For generating the desired waveform, the radar modulator makes the transmitting tube ON and OFF. It allows for storing the energy in a capacitor bank during rest time. The stored energy is kept in the pulse when it is transmitted. Commonly a thyratron valve is being used as a switching device which is a gas-filled valve.

Trigger source (Synchroniser):
A Trigger source (synchroniser) is used to regulate the rate at which the pulses are sent. It sets the PRF and resets the timing clock for the determination of the range for each pulse.
Signals from the trigger source or synchronizer are sent simultaneously to the transmitter and to the display. The transmitter sends a new pulse after receiving the signal from the trigger source, and the display resets the return sweep after receiving the signal from the trigger source.

Grey Matter Alert!

Benefits of a Radar are:
i) It can identify whether the object is stationary or moving.
ii) It does not require a medium of transportation.
iii) It allows 3-Dimensional imaging depending on the various angles of return.
iv) It can target several objects simultaneously.

Receiver:
A superheterodyne type of receiver is used to detect the desired echo signal in the presence of interference, noise, and clutter. In pulsed radar, the receiver consists of the low noise RF amplifier, mixer, local oscillator, IF amplifier, detector, video amplifier, and the radar display.

Low noise RF amplifier:
The first stage of the receiver is considered as low noise RF amplifier. It is also referred to as low noise transistor amplifier or a parametric amplifier or a TWT amplifier. For lower radar frequencies (i.e., below L-band 1215 to 1400 MHz) silicon bipolar transistor is used and for higher frequencies GaAs FET is preferred.

Mixer and local oscillator:
A mixer and a local oscillator are used to convert a RF signal output from the RF amplifier to lower frequency levels known as an intermediate frequency (IF). For a pulse radar, the value of IF is 30 MHz or 60 MHz.

IF amplifier:
It consists of a cascaded form of tuned amplifiers. All these individual amplifiers are tuned to the same frequency and have an identical bandpass characteristics. For a pulse radar, the typical value of frequency of the IF amplifier is around 30 MHz or 60 MHz.

Detector and video amplifier:
The detector used is a Schottky-barrier diode detector. For extracting the pulse modulation from the IF amplifier output, this detector is used. The output which is detected by the detector is amplified by the video amplifier at a level where it can be clearly displayed on a CRT (Cathode Ray Tube) directly or with the help of digital signal processors. Synchronising pulses are applied by the trigger source to the display unit.

Definitions

"The ability of a Radar to differentiate between two or more targets that are very close to each other is defined as the range resolution".

Display unit:
The received video signals are displayed on the CRT (Cathode-Rays Tube) for further observations and actions.

APPLICATIONS OF RADAR

RADAR can be used to detect targets on the ground, in the sea, in the air, in space, and even below ground. The major areas of radar applications are described below:

1) **Military applications:**
   In the military, RADAR has wide applications. It is used by the military, especially for surveillance purposes of enemy aircrafts. The surveillance RADAR can detect, locate, and identify the enemy targets. It is also being used for providing navigational aids to the military as well as civil aircraft. It is also used for tracking targets and controlling weapons.
2) **Targeting radars:**
   For scanning a smaller area, the targeting radars are used. Some targeting radars have a range gate that can track a target, to eliminate the clutter and the electronic countermeasures.
3) **a) Marine navigation:**
   Marine radars are used to enhance the safety of the ship travel by warning about a collision with other ships and buoys. For the surveillance of harbours, Shore-based radars are used.
   b) Aircraft navigation:
   A weather radar is used on aircraft to outline the precipitation regions to the pilot in a classical form. Weather radars are also employed in airports to warn the pilot of thunderstorms. Besides these, radar is also utilised for terrain avoidance and terrain following during aircraft navigation.

Rack Your Brain

What are the parameters on which the degree of range resolution depends?

4) **Weather radars:**
Weather radars are similar to search radars. These radars use radio waves with the polarisation type being horizontal, dual (horizontal and vertical), or circular. Some weather radar uses dopplers to measure wind speed.

5) **Remote sensing:**
The term remote sensing means the sensing of geophysical objects of the environment from a remote location. Thus, radar was used to probe the moon and the planets. Remote sensing radars borne in satellites are now used for remote sensing of earth resources, water resources, and environmental pollution.

6) **Radar altimeters:**
These are used to determine the actual height of aircraft above the ground.

7) **Air traffic control (ATC):**
RADARs are used throughout the world for the safety of aircraft and for controlling air traffic. RADARs are used to guide the aircrafts for safe landing as they are used with a ground control approach (GCA) system.

8) **Law enforcement and highway safety:**
To enforce the speed limits, RADAR speed meters are used by police.

9) **Scientific applications:**
The use of radar as a measurement tool by research scientists has vastly increased our knowledge of meteorology, aurora, meteors, and other objects of the solar system. Radars are used to guide space vehicles and satellites and may be used for the exploration of interplanetary space.

Definitions

"In a Radar system, if the target is moving, then the change in frequency of a signal takes place which is received by the Radar. This effect is referred to as Doppler effect".

RADAR FREQUENCY BANDS

The various operating frequencies for Radar applications are as shown in the below table:

Band	Frequency Range
HF	3 – 30 MHz
VHF	30 – 300 MHz
UHF	300 – 1000 MHz
L	1 – 2 GHz
S	2 – 4 GHz
C	4 – 8 GHz
X	8 – 12 GHz
Ku	12 – 18 GHz
K	18 – 26.5 GHz
Ka	26.5 – 40 GHz
mm	40 – 300 MHz

Table 9.1 Radar frequency bands

Definitions

"In a Radar system, the equation which is used to calculate the theoretical range of the target is termed as Radar range equation".

RESOLUTION

It is defined as the ability of RADAR to separate and detect multiple targets or multiple features on the same target. In other words, a radar can distinguish targets that are very close in either range or bearing.

Range resolution (RS):
RADAR can distinguish two or more targets at different ranges but at the same bearing. It has units of distance.

Range Resolution, $RS = v_0 \frac{PW}{2}$

Where

v_0 = Velocity
PW = Pulse width

Grey Matter Alert!

An electrical engineer and a physicist Sir Charles Kuen Kao pioneered the development and the use of fibre optics in telecommunications. He is referred to as the "Father of fibre optic communications".

RADAR RANGE EQUATION

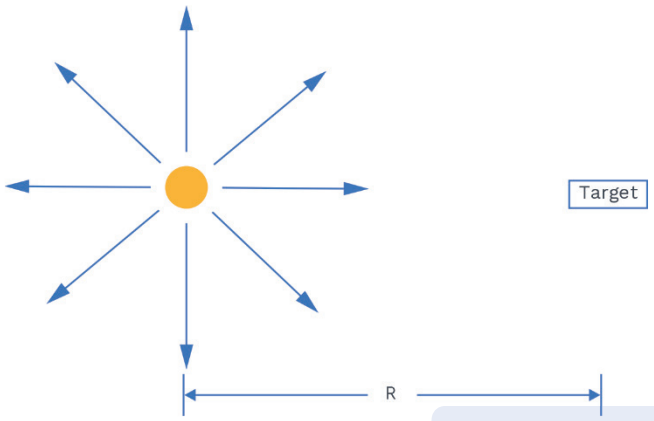


Fig. 9.6 An isotropic antenna

If the power of the radar transmitter is denoted by P_t and if there is an isotropic antenna, then the power density at a distance 'R' from the radar is given by:

$$\frac{P_t}{4\pi R^2} \text{ watt / m}^2 \qquad \text{...(i)}$$

But in RADARs a directive antenna is used (with narrow beamwidth) to concentrate the radiated power P_t in a particular direction.
Power density at a distance R from the directive antenna of power Gain 'G' is given by:

$$\frac{P_t G}{4\pi R^2} \text{ watt / m}^2 \qquad \text{...(ii)}$$

Only a small fraction of the incident energy is intercepted by the target, which re-radiates it in numerous directions. It is only the power density re-radiated in the direction of the radar, i.e., echo signal, Which is of use the radar cross-section of the target (denoted by σ) determines the power density returned to the radar for a particular power density incident on the target. It is called as target cross-section or Radar Cross Section.
The re-radiated power density back at the radar

$$= \frac{P_t G \sigma}{\left(4\pi R^2\right)\left(4\pi R^2\right)} \qquad \text{...(iii)}$$

Definitions

"A cylindrical waveguide made of transparent dielectric such as glass or clear plastic, which guides the light waves along its length by total internal reflection is referred to as optical fibre".

The radar cross-section basically denotes area. So, its unit is the same as that of the area.
The radar antenna receives only a small portion of the echo power.
If the effective area of receiving antenna is denoted A_e, the power received by radar is denoted by P_r, then:

$$P_r = \frac{P_t G \sigma A_e}{(4\pi R^2)(4\pi R^2)} \text{ watts} \qquad ...(iv)$$

The maximum range of radar R_{max} is defined as the distance beyond which the targets cannot be detected.
It occurs only when the received signal power P_r just equals the minimum detectable signal (S_{min}).
Hence, $S_{min} = P_r$

$$S_{min} = \frac{P_t G \sigma A_e}{(4\pi R^2)(4\pi R^2)} \text{ watts} \qquad ...(v)$$

$$\text{i.e., } R_{max}^4 = \frac{P_t G \sigma A_e}{(4\pi)^2 (S_{min})} \qquad ...(vi)$$

$$R_{max} = \left(\frac{P_t G \sigma A_e}{(4\pi)^2 S_{min}} \right)^{1/4} \qquad ...(vii)$$

Where

P_t = Transmitted Power
G = Maximum Gain of Antenna
A_e = Aperture area of receiving Antenna
σ = Cross-section area of the target
S_{min} = Minimum Detectable Signal
R_{max} = Maximum range of Radar

Hence, the maximum radar range is directly proportional to one-fourth power of antenna gain. This equation (vii) is referred to as Radar Range Equation. This is the fundamental form of the radar range equation.

Rack Your Brain

What is the working principle of optical fibre communication?

Definitions

"In an optical fibre, the light carrying portion of the fibre is termed as the core".

We know that the relationship between transmitting gain G and the receiver effective area A_e is:

$$G = \frac{4\pi A_e}{\lambda^2} \qquad \ldots(viii)$$

Where λ = wavelength

$$\left(\lambda = \frac{c}{f} \text{ where c = velocity of wave and f = frequency} \right)$$

Now from eq. (vii) and (viii) we get

$$R_{max} = \left(\frac{P_t G^2 \sigma \lambda^2}{(4\pi)^3 S_{min}} \right)^{1/4} \qquad \ldots(ix)$$

And

$$R_{max} = \left(\frac{P_t A_e^2 \sigma}{4\pi \lambda^2 S_{min}} \right)^{1/4} \qquad \ldots(x)$$

These three forms of the radar equation (vii), (ix), and (x) are the same.

Definitions

In an optical fibre, the layer of material which surrounds the core is termed as cladding".

Example 6:

For transmission and reception, a radar uses a common antenna which is operating at 3 GHz. The gain of antenna is 100, that is aligned for the maximum directional radiation and reception to a target of 2 km away with a radar cross section of 4 m^2. If it transmits the power of 400 kW, then obtain the received power (in nW).

Sol:

The correct answer is 5.039

Given,

Frequency, f = 3 GHz

Gain, G = 100

Radar cross-section, σ = 4 m^2

Transmitted power, P_t = 400 kW

Target distance, R = 2 km

Received power in radar is given by

$$S_{min} = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4}$$

$$= \frac{400 \times 10^3 \times (100)^2 \times \left(\frac{3 \times 10^8}{3 \times 10^9}\right)^2 \times 4}{(4\pi)^3 \times (2 \times 10^3)^4}$$

$$= \frac{400 \times 10^7 \times (0.1)^2 \times 4}{(4\pi)^3 \times 1.6 \times 10^{13}}$$

= 5.039 nW

Hence, the correct answer is 5.039



Rack Your Brain

Which material is used for the fabrication of optical fibres in communication?

OPTICAL FIBRE FUNDAMENTALS

An optical fibre is a thin transparent flexible strand (wire) that consists of a core surrounded by cladding. It confines electromagnetic energy in the form of light within its surfaces and guides the light in a direction parallel to its axis. The core is the portion of the fibre that carries the transmitted light. The cladding is used to surround the core. It has a lower index of refraction to keep the light in the core. Optical fibre has an additional coating around the cladding called a protective jacket. It protects the core and cladding from shocks that might affect their optical or physical properties. It has no optical properties affecting the propagation of light within the fibre.

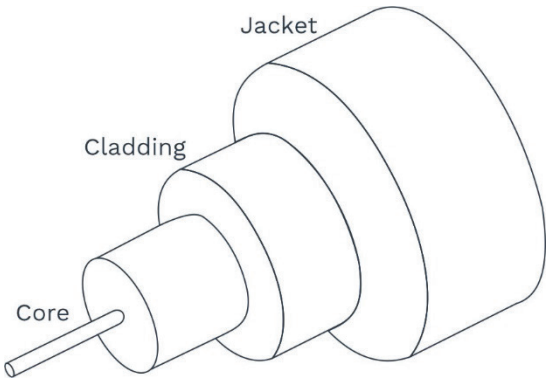


Fig. 9.7 Basic arrangement of an optical fibre

Grey Matter Alert!

The advantages of fibre optics in data transmission are:

- high bandwidth
- high speed
- high flexibility
- low signal loss

GENERAL OPTICAL FIBRE COMMUNICATION SYSTEM

Basic block diagram of an optical fibre communication system as shown in the figure below consists of:

i) Transmitter
ii) Information channel
iii) Receiver

Rack Your Brain

Explain the limitations of optical fibre in communication.

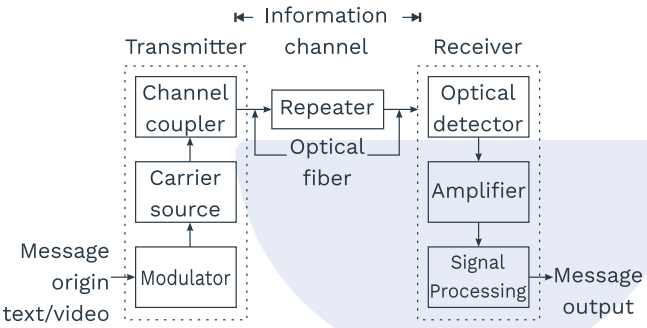


Fig. 9.8 Block diagram of OFC systems

1) **Message origin:** A message origins from a transducer that converts a non-electrical message signal into an electrical signal. For example: Microphones for converting the sound waves into current and video (TV) cameras for converting the images into the current.
2) **Modulator:** The two main functions of a modulator are:
   i) It converts the electrical message into the proper format.
   ii) It impresses this signal onto the wave generated by the carrier source.
   Two distinct categories of modulation are used, i.e., analog us modulation and digital modulation.
3) **Carrier source:** The carrier source generates the wave on which the information is transmitted. This wave is called the carrier. For a fibre optic system, a laser diode (LD) or a light-emitting diode (LED) is used. They can be called optic oscillators. They provide stable, single-frequency waves with sufficient power for long-distance propagation.
4) **Channel coupler:** The coupler feeds the power into the information channel. For an atmospheric optic system, the channel coupler is a lens used for collimating the light emitted by the source and directing this light towards the receiver. The coupler must efficiently transfer the modulated light beam from the source to the optic fibre. The channel coupler design is an important part of the fibre system because of the possibility of high losses.

5) **Information channel:** The information channel is the path between the transmitter and receiver. In fibre-optic communications, a glass or plastic fibre is the channel. Desirable characteristics of the information channel include low attenuation and large light acceptance cone angle. Optical amplifiers boost the power levels of weak signals. Amplifiers are needed in very long links to provide sufficient power to the receiver. Repeaters can be used only for digital systems. They convert weak and distorted optical signals to electrical ones and then regenerate the original digital pulse trains for further Transmission.
Another important property of the information channel is the propagation time of the waves travelling along with it. A signal propagating along with a fibre normally contains a range of optic frequencies and divides its power along several ray paths. This results in a distortion of the propagating signal. In a digital system, this distortion appears as a spreading and deforming of the pulses. The spreading is so great that adjacent pulses begin to overlap and become unrecognisable as separate bits of information.
6) **Optical detector:** The information being transmitted is detected by the detector. In the fibre system, the optic wave is converted into an electric current by a photodetector. The current developed by the detector is proportional to the power in the incident optic wave. Detector output current contains the transmitted information. This detector output is then filtered to remove the constant bias and then amplified.



Rack Your Brain

Explain why fibre optic is not suitable for broadcasting?

The important properties of photodetectors are small size, economy, long life, low power consumption, high sensitivity to optic signals, and fast response to quick variations in the optic power.

7) **Signal processing:** Signal processing includes filtering, amplification. Proper filtering maximises the ratio of signal to unwanted power. For a digital system, the decision circuit is an additional block. The bit error rate (BER) should be very small for quality communications.
8) **Message output:** The electrical form of the message emerging from the signal processor is transformed into a sound wave or visual image. Sometimes these signals are directly usable when computers or other machines are connected through a fibre system.

ADVANTAGES OF OPTICAL FIBRE COMMUNICATIONS

1) **Wide bandwidth:** The light wave occupies the frequency range between $2 \times l0^{12}$ Hz to 37×10^{12}Hz. Thus, the information-carrying capability of fibre optic cables is much higher.
2) **Low losses:** Fibre optic cables offer very less signal attenuation over long distances. Typically, it is less than 1dB/km.
3) **Immune to cross talk:** Fibre optic cables are non-conductors of electricity. Hence, they do not produce a magnetic field. Thus fibre optic cables are immune to cross-talk between cables caused by magnetic induction.
4) **Interference immune:** Fibre optic cables are immune to conductive and radiative interferences caused by electrical noise sources such as lightning, electric motors, fluorescent lights.
5) **Lightweight:** Fibre cables are made of silica glass or plastic, which is much lighter than copper or aluminium cables. Lightweight fibre cables are cheaper to transport.

Rack Your Brain

What is the range of wavelength (λ) of electromagnetic spectrum which is used by optical fibre communication?

6) **Small size:** The diameter of the fibre is much smaller compared to other cables. Therefore, fibre cable is small in size and requires less storage space.
7) **More strength:** Fibre cables are stronger and rugged and hence can support more weight.
8) **Security:** Fibre cables are more secure than other cables. It is almost impossible to tap into a fibre cable as they do radiate signals. No ground loops exist between optical fibres and hence they are more secure.
9) **Long distance transmission:** Because of less attenuation, transmission at a longer distance is possible.
10) **Environment immune:** Fibre cables are more immune to environmental extremes. They can operate over large temperature variations. Also, they are not affected by corrosive liquids and gases.
11) **Safe and easy installation:** Fibre cables are safer and easier to install and maintain. They are nonconductors. Hence, there are no shock hazards as no current or voltage is associated with them. Their small size and lightweight feature make installation easier.
12) **Less cost:** The cost of an optical fibre system is less compared to any other system.

DISADVANTAGES OF OPTICAL FIBRE COMMUNICATIONS

The disadvantages of optical fibres are as follows:

1) **Strength:** Optical fibres have a significantly lower tensile strength than coaxial cables. This can be improved by providing a protective jacket of PVC. In addition, glass fibre is much more fragile than copper wire, making fibre less attractive where hardware portability is required.



Rack Your Brain

The bandwidth of an optical fibre is:
a) 900 Hz
b) 900 MHz
c) 900 PHz
d) 900 THz

2) **Optical fibre cables are more susceptible to losses introduced by bending the cable:** Electromagnetic waves propagate through an optical cable by either refraction or reflection. Therefore, bending the cable causes irregularities in the cable dimensions, It results in a loss of signal power. Optical fibres are more prone to manufacturing defects, as even minor defects can cause excessive loss of signal power.
3) **Specialised tools, equipment, and training:** Optical fibre cables require special tools to splice and repair cables and special test equipment to make routine measurements. Not only repairing the fibre cables is difficult and expensive, but technicians working on optical cables also require special skills and training. Sometimes it is difficult to locate faults in optical cables because there is no electrical continuity.

Definitions

"When the ray of light travelling from the denser medium to the rarer medium at an angle greater than the critical angle, then the ray is totally reflected back to the denser medium. This phenomenon is termed as Total Internal Reflection (TIR)".

ELECTROMAGNETIC SPECTRUM

Electromagnetic Spectrum:

1) The radio waves and light are electromagnetic waves. The rate at which they alternate in polarity is called their frequency '*f*' measured in Hertz (Hz). The speed of the electromagnetic wave (c) in free space is approximately 3×10^8 m/sec. The distance travelled during each cycle is called wavelength (λ).

$$\text{Wavelength}(\lambda) = \frac{\text{Speed of light}}{\text{Frequency}} = \frac{c}{f}$$

2) In fibre optics, it is more convenient to use the wavelength of light instead of the frequency with light frequencies. Wavelength is often stated in microns or nanometers.
1 micron (μ) = 1 micrometer (1×10^{-6})
1 nano (n) = 10^{-9} meter

Rack Your Brain

The refractive index of core is:
a) higher than that of cladding
b) lower than that of cladding
c) equal to that of cladding
d) none of the above

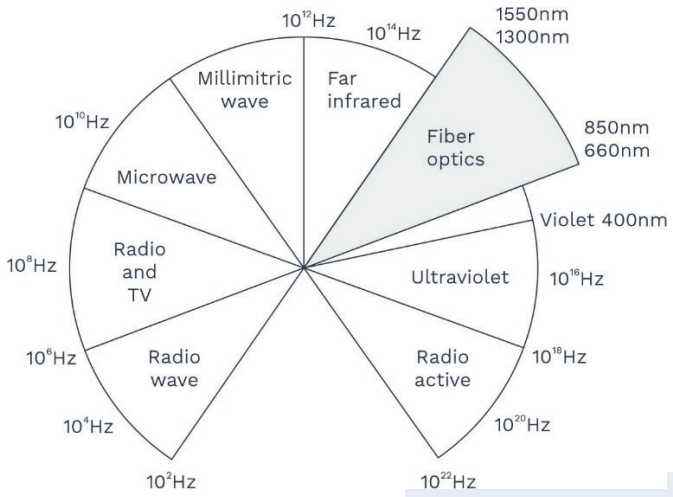


Fig. 9.9 Electromagnetic frequency spectrum.

3) Fibre optics use visible and infrared light covers a fairly wide range of wavelengths and is generally used for all fibre optics communications. Visible light is normally used for short-range transmission using a plastic fibre.

Applications of Optical Fibre Communication System

Fibre optic cables find many uses in a wide variety of industries and applications. Some uses of fibre optic cables include:

1) **Medical:** Used as light guides, imaging tools, and also as lasers for surgeries.
2) **Defense/Government:** Used as hydrophones for seismic and SONAR uses, as wiring in aircraft, submarines, and other vehicles, and also for field networking.
3) **Data Storage:** Used for data transmission.
4) **Telecommunications:** Fibre is laid and used for transmitting and receiving purposes.
5) **Networking:** Used to connect users and servers in a variety of network settings and help to increase the speed and accuracy of data transmission.

Grey Matter Alert!

Most of the optical fibres used in communication has a diameter of 0.25 mm-0.5 mm (including the outer coating).

6) **Industrial/Commercial:** Used for imaging in hard to reach areas, such as wiring where EMI is an issue, as sensory devices to make temperature, pressure, and other measurements, and as wiring in automobiles and in industrial settings
7) **Broadcast/CA TV:** Broadcast/cable companies are using fibre optic cables for wiring CATV, HDTV, internet, video-on-demand, and other applications.

RAY TRANSMISSION THEORY

Total internal reflection:

The refractive index of a medium is defined as the ratio of the velocity of light in a vacuum to the velocity of light in the medium. A ray of light travels more slowly in an optically dense medium than in one that is less dense, and the refractive index gives a measure of this effect. When a ray is incident on the interface between two dielectrics of differing refractive indices (e.g., glass-air), refraction occurs, as illustrated in figure 1 (a). It may be observed that the ray approaching the interface is propagating in a dielectric of refractive index n_1 and is at an angle ϕ_1 to the normal at the surface of the interface. If the dielectric on the other side of the interface has a refractive index n_2, which is less than n_1, then the refraction is such that the ray path in this lower index medium is at an angle ϕ_2 to the normal, where ϕ_2 is greater than ϕ_1. The angle of incidence is ϕ_1 and refraction is ϕ_2 to the normal, where ϕ_2 is greater than ϕ_1. The angles of incidence ϕ_1, and refraction ϕ_2 are related to each other and to the refractive indices of the dielectric by Snell's law of refraction, which states that: $n_1 \sin\phi_1 = n_2 \sin\phi_2$

Grey Matter Alert!

Optical fibres have negligible loss in propagation, but some losses are still present, which are as follows:

- Material absorption
- Linear and non-linear scattering
- Fibre band losses

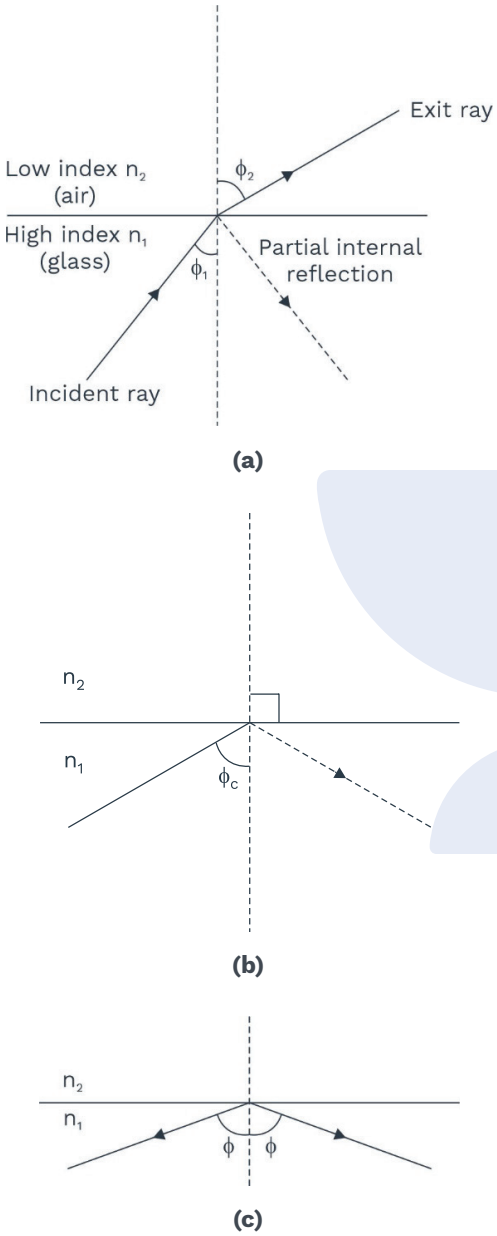


Fig. 9.10 Light rays incident on a high to low refractive index interface (e.g. glass-air) (a) refraction; (b) the limiting case of refraction showing the critical ray at an angle ϕ_c. (c) total internal reflection where $\phi > \phi_c$.

or $$\frac{\sin\phi_1}{\sin\phi_2} = \frac{n_2}{n_1} \quad \text{...(i)}$$

Rack Your Brain

If the angle of acceptance is 15 degrees, then the numerical aperture of a fibre is:

a) 0.17 **b)** 0.26
c) 0.5 **d)** 0.75

In figure 1 (a), a small amount of light is reflected into the originating dielectric medium (partial internal reflection). As n_1 is greater than n_2 the angle of refraction is always greater than the angle of incidence. Thus, when the angle of refraction is 90° and the refracted ray emerges parallel to the interface between the dielectrics, the angle of incidence must be less than 90°. This is the limiting case of refraction, and the angle of incidence is now known as the critical angle ϕ_c as shown in Figure 1 (b). From Eq. (i), the value of the critical angle is given by:

$$\sin\phi_c = \frac{n_2}{n_1} \qquad \text{...(ii)}$$

At angles of incidence greater than the critical angle, the light is reflected into the originating dielectric medium (total internal reflection) with high efficiency (around 99.9%). Hence, it may be observed in figure 1 (c) that total internal reflection occurs at the interface between two dielectrics of differing refractive indices when light is incident on the dielectric of the lower index from the dielectric of the higher index, and the angle of incidence of the ray exceeds the critical value. This is the mechanism by which light at a sufficiently shallow angle (less than 90° $-\phi_2$) may be considered to propagate down an optical fibre with low loss.

Grey Matter Alert!

Attenuation in optical fibres is caused due to:

- Absorption
- Scattering
- Waveguide effect

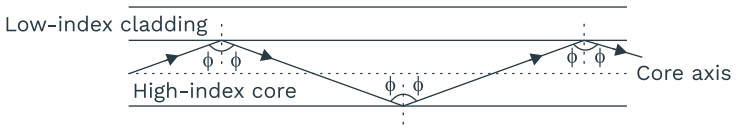


Fig. 9.11 The transmission of a light ray in a perfect optical fibre

Figure 2 illustrates the transmission of a light ray in an optical fibre via a series of total internal reflections at the interface of the silica core and the slightly lower refractive index silica cladding.

The ray has an angle of incidence ϕ at the interface, which is greater than the critical angle and is reflected at the same angle as the normal. The light ray shown in figure 2 is known as a meridional ray as it passes through the axis of the fibre core.

Acceptance angle:

The geometry concerned with launching a light ray into an optical fibre is shown in figure 3, which illustrates a meridional ray A at the critical angle ϕ_c within the fibre at the core-cladding interface. This ray enters the fibre core at an angle θ_a to the fibre axis and is refracted at the air-core interface before transmission to the fibre core-cladding interface at the critical angle. Hence, any rays which are incident into the fibre core at an angle greater than θ_a will be transmitted to the core-cladding interface at an angle less than θ_c and will not be totally internally reflected. This situation is also illustrated in figure 3, where the incident ray B at an angle greater than θ_a is refracted into the cladding and eventually lost by radiation. Thus, for rays to be transmitted by total internal reflection within the fibre core they must be incident on the fibre core within an acceptance cone defined by the conical half angle θ_a. Hence, θ_a is the maximum angle to the axis at which light may enter the fibre in order to be propagated and is often referred to as the acceptance angle for the fibre.

Grey Matter Alert!

Based on the modes of propagation, optical fibres are of two types:

- Single Mode Fibres (SMF)
- Multi-Mode Fibres (MMF)

Grey Matter Alert!

Single Mode Fibres are used for long distance communication, while Multi-Mode Fibres are used for short distance communication.

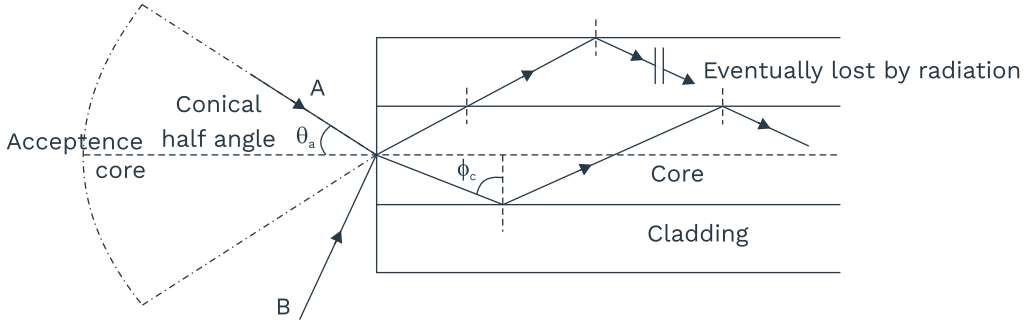


Fig. 9.12 The acceptance angle θ_a when launching light into an optical fibre

Numerical aperture:

To obtain a relationship between the acceptance angle and the refractive indices of the three media involved, namely the core, cladding, and air. Figure 4 shows a light ray incident on the fibre core at angle θ_1 to the fibre axis, which is less than the acceptance angle for the fibre θ_a.

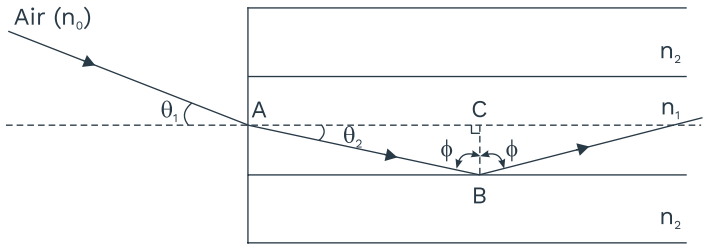


Fig. 9.13 The ray path for a meridional ray launched into an optical fibre in the air at an input angle less than the acceptance angle to the fibre

The rays enter fibre from a medium (air) of refractive index n_0 and the fibre core has a refractive index n_1 which is slightly greater than the cladding refractive index n_2. Assuming the entrance face at the fibre core to be normal to the a_x is, then considering the refraction at the air-core interface and using Snell's law given by equation (i).

$$n_0 \sin\theta_1 = n_1 \sin\theta_2 \qquad \text{...(iii)}$$

Considering the right-angled triangle ABC indicated in figure 4, then:

$$\phi = \frac{\pi}{2} - \theta_2 \qquad \text{...(iv)}$$

where ϕ is greater than the critical angle at the core-cladding interface. Hence equation (iii) becomes:

$$n_0 \sin\theta_1 = n_1 \cos\theta_2 \qquad \text{...(v)}$$

Using the trigonometrically relationship $\sin^2\phi + \cos^2\phi = 1$, equation (v) may be written in the from:

$$n_0 \sin\theta_1 = n_1 (1 - \sin^2\phi)^{1/2} \qquad \text{...(vi)}$$

When the limiting case for total internal reflection is considered, ϕ becomes equal to the critical angle for the core-cladding interface and is given by equation (ii). Also, in this limiting case, θ_1 becomes the acceptance angle for the fibre θ_a. Combining these limiting cases into equation (vi) gives:

$$n_0 \sin\theta_a = (n_1^2 - n_2^2)^{1/2} \qquad \text{...(vii)}$$

Equation (vii), apart from relating the acceptance angle to the refractive indices, serves as the basis for the definition of the important optical fibre parameter, the numerical aperture (NA). Hence, NA is defined as.

$$NA = n_0 \sin\theta_a = (n_1^2 - n_2^{2})^{1/2} \quad \text{...(viii)}$$

Since NA is often used with the fibre in the air where n_0 is unity. It is simply equal to sin θ_a. It may also be noted that incident meridional rays over the range $0 \le \theta_1 \le \theta_a$ will be propagated within the fibre.

The NA may also be given in terms of the relative refractive index difference Δ between the core and the cladding which is defined as:

$$\Delta = \frac{n_1^2 - n_2^2}{2n_1^2} = \frac{n_1 - n_2}{n_1} \text{ for } \Delta << 1 \quad \text{...(ix)}$$

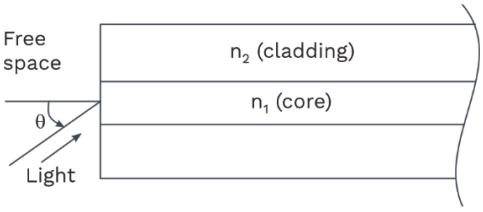
Hence combining equation (viii) with equation (ix) we can write:

$$NA = n_1\sqrt{2\Delta} \quad \text{...(x)}$$

Sometimes another parameter $\Delta n = n_1 - n_2$ is referred to as the index difference and $\Delta n/n_1$ as the fractional index difference, Hence, Δ also approximates to the fractional index difference.

Example 7:

Light from free space is incident at an angle θ_i to the normal of the facet of step-index large core optical fibre. The core and cladding refractive indices are $n_1 = 1.384$ and $n_2 = 1.201$, respectively.



The maximum value of θ_i (in degrees) for which the incident light will be guided in the core of the fibre is _____.

Sol:

The correct answer is 43.45

Given,

Refractive index of core, $n_1 = 1.384$

Refractive index of cladding, $n_2 = 1.201$

Angle of Acceptance = θ_i

Angle of Acceptance is given by formula,

$\theta_i = \sin^{-1}(NA)$

Where

NA = Numerical aperture $= \sqrt{n_1^2 - n_2^2}$

$= \sqrt{1.384^2 - 1.201^2}$

= 0.6877

Therefore, angle of acceptance will be,

$\theta_i = \sin^{-1}(NA) = \sin^{-1}(0.6877)$

$\theta_i = 43.45°$

Hence, the correct answer is 43.45

Example 8:

A typical relative refractive index difference for an optical fibre defined for long-distance transmission is 5%, when the core index is 1.516.

The numerical aperture is _____.

Sol:

The correct answer is 0.479

Given,

Relative refractive index,

$\Delta = 5\% = 0.05$

Core refractive index, $n_1 = 1.516$

Numerical aperture in terms of relative refractive index is given by,

$NA = n_1\sqrt{2\Delta}$

$= 1.516\sqrt{2 \times 0.05}$

= 0.479

Hence, the correct answer is 0.479

Example 9:

A monochromatic light passes through the two glass slabs at normal incidence having the same geometrical thickness. If the refractive index of glass slab (1) and glass slab (2) are 1.75 and 2.0, respectively, then obtain the ratio of the time of passage of the beam through the first glass slab to the second glass slab.

Sol:

The correct answer is 0.875

Given,

Refractive index of first slab (n_1) = 1.75

Refractive index of second slab (n_2) = 2.0

$d_1 = d_2$ (same geometrical thickness)

$$\text{Time} = \frac{\text{Distance}}{\text{Velocity}} = \frac{d}{v}$$

Therefore, the ratio of time for 2 slabs $= \frac{t_1}{t_2}$

The Refractive index of the material is given by,

$$n = \frac{\text{Velocity of light in vaccum}}{\text{Velocity of light in medium}}$$

$$n = \frac{c}{V_P} \Rightarrow V_P = \frac{c}{n}$$

Now, $t_1 = \frac{l}{V_{P_1}}$ and $V_{P_1} = \frac{c}{n_1}$

$$t_2 = \frac{l}{V_{P_2}} \text{ and } V_{P_2} = \frac{c}{n_2}$$

Then, $\frac{t_1}{t_2} = \frac{\frac{l \cdot n_1}{c}}{\frac{l \cdot n_2}{c}} = \frac{n_1}{n_2} = \frac{1.75}{2.0} = 0.875$

Hence, the ratio of time passage between the first and second slab is 0.875.
Hence, the correct answer is 0.875

Example 10:
Light pulse propagates through a fibre cable with an attenuation of 0.625 dB/km. The distance through which the power of pulses is reduced to 50%, is _____ km.

Sol:
The correct answer is 4.816
Given,
Attenuation constant of optical fibre = 0.625 dB/km
Let the signal strength at length 'l' be = P(l)
Attenuation in optical fibre is given by

$$\alpha \times l = -10 \log_{10}\left[\frac{P(l)}{P(0)}\right]$$

$$0.625 \times l = -10 \log_{10}\left[\frac{0.5P(0)}{P(0)}\right]$$

{∵ Strength reduces to 50%}
l = 4.816 km
Hence, the distance at which power of pulses is reduced to 50% is 4.816 km.
Hence, the correct answer is 4.816

Chapter Summary



- Range of a Radar target is given as:

$$R = \frac{cT_R}{2} = \frac{c \times PRT}{2}$$

Where

c = speed of wave (velocity of light)

T_R = PRT = time taken for wave to return back to radar

- Duty cycle of radar pulse is given by:

$$\text{Duty Cycle} = \frac{PW}{PRT}$$

Where PW = pulse width

PRT = pulse repetition time

- Average Power:

$P_{av} = P_t \times$ Duty cycle

Where

P_t = power transmitted

- Maximum Unambiguous Range:

$$R_{max} = \left(\frac{P_t G \sigma A_e}{(4\pi)^2 S_{min}} \right)^{\frac{1}{4}}$$

- Snell's law:

$n_1 \sin\theta_1 = n_2 \sin\theta_2$

- Acceptance angle of fibre is:

$$\sin\theta_A = \sqrt{n_1^2 - n_2^2}$$

- Numerical aperture is given by:

$$NA = \sqrt{n_1^2 - n_2^2}$$

- Relative refractive index is given as:

$$\Delta = \frac{n_1^2 - n_2^2}{2n_1^2}$$

- V-number:

$$V = \frac{2\pi}{\lambda} \times a \times NA$$

- Number of modes:

$$M = \frac{V^2}{2}$$

PRACTICE QUESTIONS

Q1 **For a pulsed radar, if keeping the radar parameters unchanged, the peak radar transmitter power is increased by a factor of 81. By what factor the maximum range capability of the radar increases?**

a) 81 **b) 9** **c) 27** **d) 3**

Sol: The correct option is d)

Q2 **Calculate the duty cycle of a radar if the transmitter has a PRF of 1500 PPS with a pulse width of $1\mu sec$.**

a) 0.0075 **b) 0.015** **c) 0.0015** **d) 0.00075**

Sol: The correct option is d)

Q3 **The greatest allowable PRF (in pps) for an unambiguous reception in a radar having a maximum range of 120 km is:**

a) 1000 **b) 2505** **c) 1250** **d) 1850**

Sol: The correct option is a)

Q4 **For a monostatic radar, if the maximum range is R and the radar cross-section of a target is $10m^2$, which exists at R/2, then the target cross-section (in m^2) at 3R/2, which results in the equal signal strength at the radar is:**

a) 95 **b) 270** **c) 680** **d) 810**

Sol: The correct option is c)

Q5 **By what factor the maximum range of a radar will increase if the diameter of an antenna is increased by a factor of 2, and the peak transmitted power is increased by a factor of 16?**

a) $\sqrt{8}$ **b) 4** **c) 2** **d)** $\sqrt{2}$

Sol: The correct option is b)

Q6 **What is the effect on the peak transmitted power of the radar if the range is doubled?**

a) Increased by a factor of 2 **b) Increased by a factor of 4**

c) Decreased by a factor of 4 **d) Increased by a factor of 16**

Sol: The correct option is d)

Q7 **In a monostatic radar, if the antenna aperture is doubled, the radar range will:**

a) Reduce by a factor of 2 **b) Increase by a factor 2**

c) Reduce by a factor of $\sqrt{2}$ **d) Increase by a factor of** $\sqrt{2}$

Sol: The correct option is d)

Q8 **If the average power of the radar transmitter is 2kW and the peak power of the transmitter is 1000 kW, then what will be the duty cycle?**

a) 0.002 **b) 0.005** **c) 0.0002** **d) 0.1**

Sol: The correct option is a)

Q9 **In a step-index fibre the relative permittivity of a cladding material is 2.4375. It is to be assumed that both the core and cladding materials are non-magnetic, to have an index numerical aperture of 0.25, the refractive index of the core should be:**

a) 5 **b) 2.5** **c) $\sqrt{3.5}$** **d) $\sqrt{2.5}$**

Sol: The correct option is d)

Q10 **The cut-off wavelength (in μm) for an optical fibre having a diameter of 3 μm with a core index of 1.545 and a cladding index of 1.510 is:**

a) 2.10 **b) 3.05** **c) 1.29** **d) 0.941**

Sol: The correct option is c)